

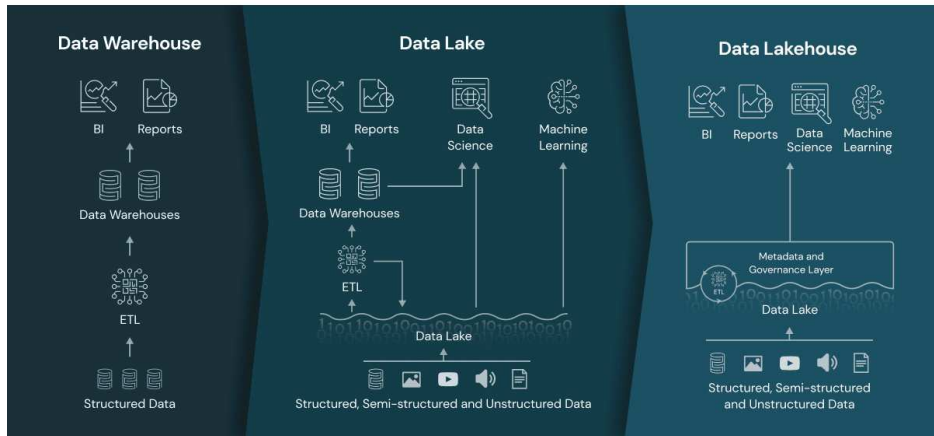
# Open table format and Iceberg

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# Evolution



Source: <https://www.databricks.com/glossary/data-lakehouse>

# Data Lakehouse

- Supports all data types
- Flexible schema
- Cost-effective storage
- Real-time data processing
- Ideal for ML/AI workloads

# Open table formats

- Standards for storing tabular data, built on top of existing file formats like Parquet, ORC or Avro.
- Tabular data is stored in ordinary files.
- Besides data itself, also metadata about the tabular data is stored in other files/directories.
- Open table format is basically a standard of this metadata layer.
- This metadata layer allows to work with a set of ordinary Parquet/ORC files like with a database table and query it.

# Open table formats features

- Fast queries.
- ADIC transactions.
- Schema evolution.
- Support for streaming I/O (ingestion of streaming data).
- Time travel.

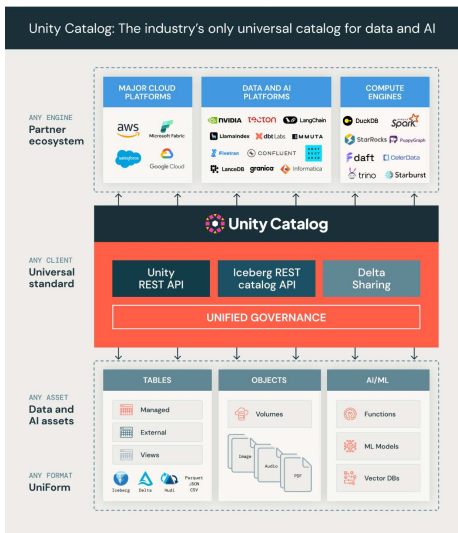
# Open table formats

Main open table formats:

- Delta Lake
- Apache Hudi
- Apache Iceberg

# Open table format unification

Attempts to unify (not only) open table formats, e.g. [Unity catalog](#)(Databricks)



Source: <https://github.com/unitycatalog/unitycatalog>

- One of the open table formats.
- Developed by Netflix and open-sourced in 2017.
- Supported by all major query/data processing engines like Spark, Trino, Flink, Hive etc. (including IBM watsonx.data).
- Possible winner in open table formats adoption.



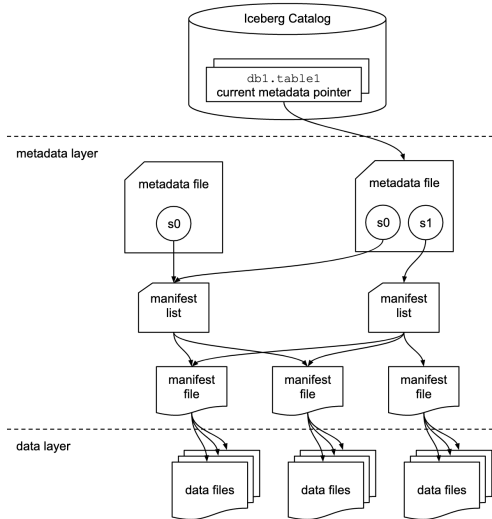
- Metadata layer between storage and compute (query engines).
- Takes snapshots of data in particular point in time - created meta-meta data.
- Manifests describe the data in Parquet files on the blob storage.
- Metadata describe the data (schema, layout, location etc.) in one particular point in time.
- Iceberg catalog tracks metadata.

# High level view

|                |         |         |        |      |
|----------------|---------|---------|--------|------|
| Compute Engine | Spark   | Flink   | Presto | Hive |
| Table format   | ICEBERG |         |        |      |
| File format    | Avro    | Parquet | ORC    |      |
| Storage        | AWS S3  | ADLS    | GCS    | HDFS |

Source: <https://medium.com/geekculture/open-table-formats-delta-iceberg-hudi-732f682ec0bb>

# Iceberg architecture



Source: <https://iceberg.apache.org/spec/>

- Catalog is top-level layer which handles queries, table creation or other DDLs.
- Many implementation, written in various languages and using various backends (files, databases, REST).
- REST catalog specifications - allow easy implementations in various languages.
- Iceberg deployment can use one or many catalogs.

# Iceberg catalog implementation (incomplete list)

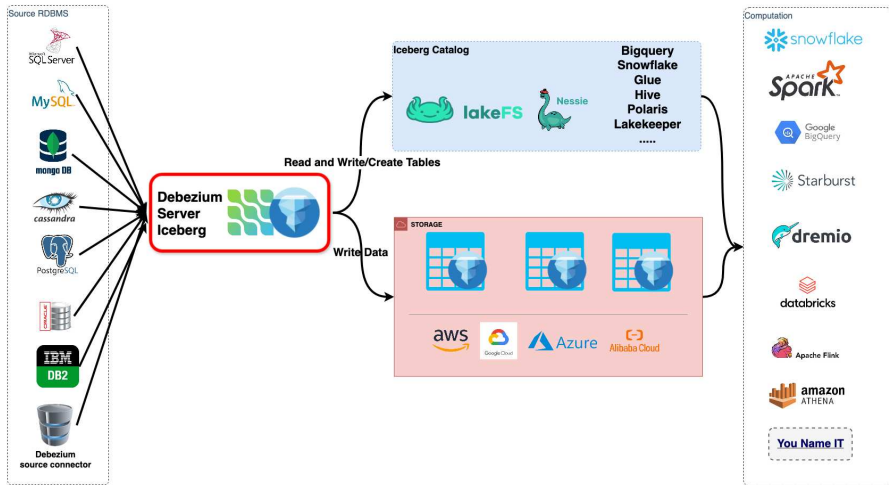
- [Nessie](#)
  - Git-inspired data version control (branches, merge of branches etc).
  - Build on top of Quarkus.
- [Apache Polaris](#)
- [Unity](#)
- [Lakekeeper](#)
- [AWS Glue](#)
- ...

# Query engines integration (incomplete list)

- Apache Spark
- Apache Flink
- Trino
- Apache Hive
- Snowflake
- StartRocks
- Google BigQuery
- ...

# Demo - Iceberg 101

# Debezium server Iceberg sink



Source: <https://github.com/memiiso/debezium-server-iceberg>



# Demo - Debezium server Iceberg sink

# Thank you!

# Questions?